

Daniel Cohen

Mathematics & Statistics Professional | Data Analyst | Quantitative Educator
GitHub: github.com/Coakley11

PROFESSIONAL SUMMARY

Quantitative professional with graduate training in statistics and finance, extensive mathematics teaching experience, and a portfolio of production analytics applications. Translates statistical theory, SQL reporting, and experimental design into interactive decision-support products for finance, sports, and AI evaluation workflows.

EDUCATION

M.S. Statistics - Hunter College

Graduate training in statistical inference, modeling, and data analysis

Finance, investment analysis, and quantitative decision frameworks

Applied mathematics, economics, and quantitative reasoning

MBA Finance

B.A. Applied Mathematics

EXPERIENCE

Analytics & AI Portfolio Developer (2024 - Present)

Independent - Daniel AI Suite

- Built and deployed seven production Streamlit applications spanning portfolio analytics, sports modeling, AI evaluation, and decision-support simulations
- Authored SQL & Excel analytics workbooks with KPI dashboards, pivot exercises, and structured query practice for recruiter-facing evidence
- Designed portfolio demo modes, case-study narratives, and QA workflows for recruiter-ready product demonstration

College Mathematics & Statistics Instructor

Higher Education

- Taught undergraduate mathematics and statistics with emphasis on applied problem solving and statistical literacy
- Developed instructional materials connecting theory to real-world quantitative decisions
- Supported student mastery of hypothesis testing, regression concepts, and data interpretation

High School Mathematics Teacher

Secondary Education

- Delivered algebra, geometry, and quantitative reasoning curriculum to diverse student populations
- Built foundational skills in logical reasoning, data literacy, and mathematical communication

Academic Tutoring - Mathematics & Statistics

Tutoring Center

- Provided individualized instruction in mathematics and statistics from remedial through advanced undergraduate levels
- Strengthened quantitative education through diagnostic assessment and targeted skill development

SELECTED PROJECTS

Investment Portfolio Analyzer | Python | Streamlit | yfinance

- Built live portfolio dashboard with Sharpe/Sortino metrics, Monte Carlo simulation, efficient frontier optimization, and correlation heatmaps
- Separated analytics core from UI for maintainability; deployed on Streamlit Cloud

Baseball Analytics Explorer | Python | Streamlit | scikit-learn

- Developed MLB analytics platform with statistical significance testing, ML projections, and fantasy draft decision support
- Implemented pytest coverage for projection calibration and draft strategy logic

Applied Mathematical Intelligence | Python | Streamlit

- Designed modular Mathematical Thinking Lab with betting EV, AI training visualization, and epidemic simulation engines
- Built pedagogical UX pattern: explain -> simulate -> interpret -> math depth

SQL & Excel Analytics Workbooks | Excel | SQL | pandas

- Created AI evaluator, investment, quant, and credit-risk workbooks with KPI dashboards and SQL practice queries
- Demonstrates fluency across Python dashboards and spreadsheet-based business reporting

TECHNICAL SKILLS

Python | SQL | Statistics | Excel | Pandas | Streamlit | Data Visualization | Git / GitHub | AI-Assisted Development | Machine Learning | Monte Carlo Simulation | Portfolio Optimization

TARGET ROLES

Data Analyst | AI Evaluator | Business Intelligence Analyst | Research Analyst | Junior Data Scientist | Quantitative Analyst